

1. Analisis Total Host Per Ruangan:

- Ruang Dosen: 30 perangkat (Wireless) → 30 host + 2 (Network & Broadcast) = 32 IP → /27
- Ruang Seminar: 8 perangkat (Wired) → 8 host + 2 = 10 IP → /28
- Lab Basis Data: 60 perangkat (Wired) → 60 host + 2 = 62 IP → /26
- Lab PPSI: 34 perangkat (Wired) → 34 host + 2 = 36 IP → /26

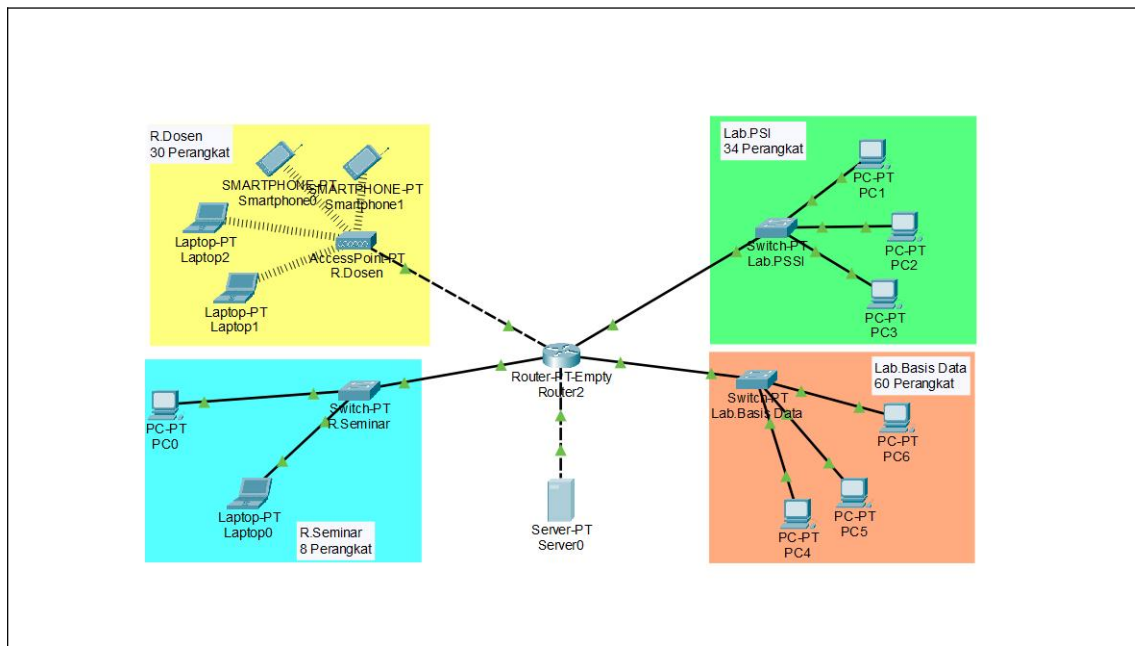
2. Pembagian Subnet (Menggunakan IP: 192.168.48.0/24)

- Lab Basis Data (/26):
 - Network: 192.168.48.0 - 192.168.48.63 (64 IP)
- Lab PPSI (/26):
 - Network: 192.168.48.64 - 192.168.48.127 (64 IP)
- Ruang Dosen (/27):
 - Network: 192.168.48.128 - 192.168.48.159 (32 IP)
- Ruang Seminar (/28):
 - Network: 192.168.48.160 - 192.168.48.175 (16 IP)
- Sisa alamat: 192.168.48.176 - 192.168.48.255 (80 IP)

3. Tabel Subnetting VLSM:

No	Subnet	Network Address	CIDR	Usable IP Range	Broadcast Address
1	Lab Basis Data	192.168.48.0	/26	192.168.48.1 - 192.168.48.62	192.168.48.63
2	Lab PPSI	192.168.48.64	/26	192.168.48.65 - 192.168.48.126	192.168.48.127
3	Ruang Dosen	192.168.48.128	/27	192.168.48.129 - 192.168.48.158	192.168.48.159
4	Ruang Seminar	192.168.48.160	/28	192.168.48.161 - 192.168.48.174	192.168.48.175
5	Sisa Alamat	192.168.48.176	/25	192.168.48.177 - 192.168.48.254	192.168.48.255

4. Bukti Topologi



5. Bukti berhasil

Fire	Last Status	Source	Destination	Type	Color	Time(sec)	Periodic	Num	Edit	Delete
	Successful	PC1	PC2	ICMP		0.000	N	0	(edit)	(delete)
	Successful	Laptop0	PC1	ICMP		0.000	N	1	(edit)	(delete)
	Successful	Laptop2	PC5	ICMP		0.000	N	2	(edit)	(delete)
	Successful	Smart...	PC3	ICMP		0.000	N	3	(edit)	(delete)